

Heterogeneous Knowledge Distillation for Real-Time 2D Hand Pose Estimation

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Abstract—Real-time hand pose estimation underpins a growing range of human-computer interaction systems, yet leading solutions are either too heavy for edge deployment or locked inside proprietary, non-retrainable pipelines such as Google’s MediaPipe Hands. This paper proposes a heterogeneous knowledge distillation framework that transfers spatial reasoning from a heatmap-based teacher (HRNetV2-W18 with DARK decoding) to a compact coordinate-regression student (BlazeHandLandmark) on the Rendered Handpose Dataset (RHD2D). The output-representation mismatch between heatmaps and coordinates is resolved by an MSRAHeatmap codec bridge that decodes teacher heatmaps into sub-pixel coordinate targets at every training iteration, supervising the student directly in pixel space without modifying either architecture and without ground-truth labels. The distilled student attains $\text{PCK@0.2}=0.748$, $\text{AUC}=0.723$, and $\text{MPJPE}=36.87$ px on the full RHD2D test set with 100% detection, while running at 139 FPS on an RTX 4050 laptop GPU—a $4.8\times$ parameter reduction and $20.2\times$ full-pipeline speedup over the teacher. The complete training pipeline is open source and retrainable, providing a transparent alternative to closed hand-tracking systems.

Index Terms—hand pose estimation, heterogeneous knowledge distillation, BlazeHandLandmark, HRNetV2-W18, real-time inference

I. INTRODUCTION

Two-dimensional hand pose estimation—the localization of 21 anatomical keypoints from a monocular RGB image—underpins gesture-based interaction, AR/VR, sign language recognition, and robotic manipulation [1], [2]. Google’s MediaPipe Hands [8] dominates real-time deployment, achieving sub-20 ms inference on mobile devices through a lightweight regression-based landmark model. However, MediaPipe is a closed system: its training data, loss functions, and model weights are proprietary and non-retrainable. Researchers cannot adapt it to specialized domains such as surgical tracking, industrial safety monitoring, or domain-specific sign language, where data distributions differ fundamentally from Google’s general-purpose training set.

Knowledge distillation (KD) [11] offers a principled path to an open alternative: train a lightweight student to mimic a high-capacity teacher, transferring spatial reasoning without the computational cost. This work applies KD to transfer knowledge from HRNetV2-W18 [3], a state-of-the-art heatmap-based teacher achieving $\text{PCK@0.2}=0.992$ on the RHD2D benchmark, into a BlazeHandLandmark student. The

result is a fully open and retrainable alternative to MediaPipe Hands that any researcher can fine-tune, inspect, and deploy on domain-specific data without dependence on proprietary infrastructure.

II. REVIEW OF RELATED WORK

Two paradigms dominate 2D hand pose estimation from monocular RGB [1]. **Heatmap-based** methods such as HRNet [3] maintain high-resolution spatial representations throughout the network, achieving state-of-the-art accuracy at computational costs that prohibit real-time edge deployment [4]. **Direct regression** methods predict coordinate vectors directly [5], enabling mobile real-time inference [6] at the cost of the spatial reasoning needed for occluded joint localization [7].

The most widely deployed regression-based solution is MediaPipe Hands [8], which combines a BlazePalm detector with a landmark regression model (1.98 M parameters, 16.1 ms on Pixel 3). Despite its practical success, its training data and pipeline are proprietary, preventing retraining or domain adaptation. The original MediaPipe paper reports only MSE normalized by palm size on a private real-world test set and does not evaluate on public benchmarks such as RHD2D [10], leaving its generalization to synthetic domains uncharacterized.

Bridging this gap requires transferring the spatial precision of heatmap-based models into a lightweight regression architecture via knowledge distillation [11]. For pose estimation, however, the key challenge is *representation heterogeneity*: heatmap teachers and regression students produce incompatible output formats. Prior work either requires ground-truth supervision alongside distillation [7] or addresses only homogeneous regressor-to-regressor transfer [13]. DARK decoding [12] recovers sub-pixel precision from heatmaps via second-order Taylor expansion but has not been applied as a GT-free distillation bridge. No existing framework performs fully heterogeneous, ground-truth-free KD from a heatmap teacher to a regression student for hand pose—the gap this work addresses.

III. PROBLEM STATEMENT AND OBJECTIVES

Despite its real-time performance, MediaPipe Hands’ proprietary training pipeline prevents researchers from retraining

or adapting it for specialized domains, motivating the need for an open-source alternative. Building one via knowledge distillation requires resolving a fundamental representation mismatch: the teacher HRNetV2-W18 with DARK decoding outputs spatial heatmaps of shape $(21 \times 64 \times 64)$, while the student BlazeHandLandmark directly regresses coordinate vectors of shape (21×3) . Prior KD work on pose estimation addresses only homogeneous settings in which teacher and student share output representations, leaving heterogeneous heatmap-to-coordinate distillation unresolved. This study bridges the gap through the MSRAHeatmap codec with second-order Taylor refinement, which converts teacher heatmaps into sub-pixel coordinate targets that directly supervise the regression student. The resulting framework yields a transparent alternative to MediaPipe Hands in which the student inherits the spatial calibration of HRNetV2-W18 while retaining real-time inference efficiency.

IV. METHODOLOGY

A. Framework Overview

Fig. 1 illustrates the proposed heterogeneous KD pipeline. A frozen HRNetV2-W18 teacher generates heatmaps that are decoded into sub-pixel coordinate targets, which then supervise the BlazeHandLandmark student via a composite Wing + Bone loss.

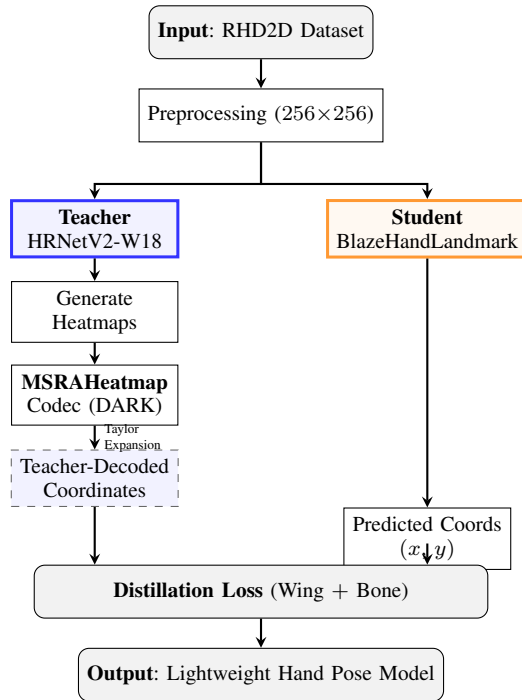


Fig. 1. Heterogeneous KD framework. The teacher (blue) decodes heatmaps via the MSRAHeatmap codec with Taylor refinement; the student (orange) directly regresses keypoints.

B. Dataset

All experiments use the **Rendered Handpose Dataset (RHD2D)** [10], a fully synthetic benchmark for single-image

hand pose estimation. Images are computer-generated from rigged 3D character meshes rendered under varied viewpoints, poses, and backgrounds, providing pixel-perfect ground-truth annotations without manual labeling [18]. Table I summarizes the dataset statistics, and Table II defines the 21-keypoint topology: one wrist joint and four joints per finger (MCP–PIP–DIP–Tip) across all five digits [14].

TABLE I
RHD2D DATASET STATISTICS

Property	Value
Training samples	41,258
Test samples	2,728
Native resolution	320×320 px
Input resolution	256×256 px
Keypoints per image	21
Annotation types	2D coords, 3D coords, visibility, hand type, segmentation mask, depth

TABLE II
RHD2D KEYPOINT DEFINITION

Index	Keypoint	Finger
0	Wrist	—
1–4	MCP–PIP–DIP–Tip	Thumb
5–8	MCP–PIP–DIP–Tip	Index
9–12	MCP–PIP–DIP–Tip	Middle
13–16	MCP–PIP–DIP–Tip	Ring
17–20	MCP–PIP–DIP–Tip	Pinky

Images are cropped via ground-truth bounding boxes, padded to a square aspect ratio, and resized to 256×256 . Training augmentation includes random horizontal flip, bounding-box scale/rotate ($\pm 180^\circ$)/shift, and photometric distortion. Performance is reported as PCK@0.2 (keypoint within $\tau = 51.2$ px of GT) and AUC (area under the PCK curve over 20 thresholds in $[0, 128]$ px) [10].

C. Distillation Loss

Let $\hat{\mathbf{p}} \in \mathbb{R}^{21 \times 2}$ and $\mathbf{p}^T \in \mathbb{R}^{21 \times 2}$ denote, respectively, the student’s predicted and the teacher-decoded keypoint coordinates. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{wing}} + \lambda \mathcal{L}_{\text{bone}}, \quad \lambda = 0.002. \quad (1)$$

Wing Loss [15] ($w=10$, $\varepsilon=2$) is logarithmic for small residuals and linear for large ones, providing robustness to outlier joints:

$$\ell_{\text{wing}}(\delta) = \begin{cases} w \ln(1 + |\delta|/\varepsilon) & |\delta| < w, \\ |\delta| - C & \text{otherwise,} \end{cases} \quad (2)$$

where $C = w(1 - \ln(1 + w/\varepsilon))$ and $\delta = \hat{p} - p^T$. The aggregate is the mean over all joints and coordinate axes:

$$\mathcal{L}_{\text{wing}} = \frac{1}{42N} \sum_{n,k,d} \ell_{\text{wing}}(\hat{p}_{n,k,d} - p_{n,k,d}^T). \quad (3)$$

Bone Loss penalizes the mean-squared error between predicted and teacher-decoded bone lengths across the 20 directed

segments $\mathcal{B}$ of the five-finger skeleton (four segments per finger, all rooted at the wrist):

$$\mathcal{L}_{\text{bone}} = \frac{1}{|\mathcal{B}|N} \sum_{(i,j) \in \mathcal{B}} \sum_n \left(\|\hat{\mathbf{p}}_{n,i} - \hat{\mathbf{p}}_{n,j}\|_2 - \|\mathbf{p}_{n,i}^T - \mathbf{p}_{n,j}^T\|_2 \right)^2, \quad (4)$$

enforcing skeletal consistency independent of absolute joint position.

D. Training Procedure

The student is trained on RHD2D in two stages using a frozen HRNetV2-W18 teacher, with hyperparameters and loss formulations summarized in Fig. 2. Both stages employ Adam optimization, cosine annealing, and the same data augmentations. Phase 1 distills from MediaPipe pretrained weights; Phase 2 introduces gradient clipping and early stopping for final refinement. Three baselines contextualize performance: a direct-supervision variant trained on GT labels (overfits after epoch 25), the HRNetV2-W18 teacher (upper-bound reference), and the official MediaPipe SDK (domain-gap reference).

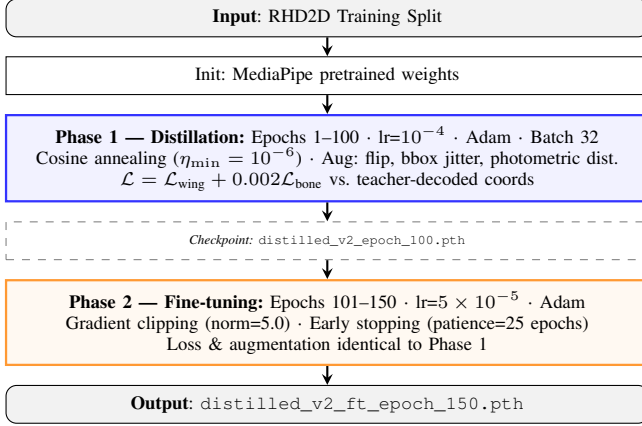


Fig. 2. Two-phase training procedure.

V. EXPERIMENTAL RESULTS

A. Evaluation Protocol

All experiments run on an NVIDIA GeForce RTX 4050 Laptop GPU (CUDA 12.1, PyTorch 2.1.0) over the full RHD2D test split (2,727 images, 21 keypoints). Three accuracy metrics are reported: PCK@0.2 ($\tau = 51.2$ px), AUC (20 thresholds in $[0, 128]$ px), and MPJPE (mean Euclidean error in pixels). GPU inference speed (FPS) is reported as the efficiency metric. All trained models are evaluated under identical conditions: affine-cropped 256×256 inputs, fixed GT bounding boxes, and the same 2,727-sample test split.

The MediaPipe Hands SDK serves as a domain-gap reference, not as a trained competitor. It is the official MediaPipe Hands application (model_complexity=1, 1.98 M parameters) evaluated directly on RHD2D to quantify the shift between its real-world training distribution and RHD2D’s flat-shaded

TABLE III
DESIGN DIFFERENCES: DISTILLATION V1 vs. V2

Component	V1	V2
Student init.	Random	BlazeHandLandmark (MediaPipe pretrained)
Augmentation	None	Flip+Scale+Rotate+Color
Loss	MSE	Wing+Bone ($\lambda=0.002$)
LR schedule	Fixed 10^{-4}	Cosine $10^{-4} \rightarrow 10^{-6}$
Validation	None	PCK+MPJPE every 5 ep.
Coord. clipping	No	$[0, 255]$
Grad. clipping	No	$\ \nabla\ _{\max} = 5.0$ (FT)
Epochs	210	100 distil. + 50 fine-tune

synthetic style [8]. Separately, the BlazeHandLandmark architecture ported to PyTorch via the zmurez repository [9] is evaluated in its *pre-distillation* state—i.e., with the original released weights, before any KD training—to establish the student’s true starting point. The two models are distinct: the MediaPipe Hands SDK operates as a full two-stage pipeline (BlazePalm detector followed by the hand landmark model), while the BlazeHandLandmark (zmurez) weights are applied directly to pre-cropped 256×256 inputs, bypassing the BlazePalm stage. Detection failures are not excluded from scoring: when BlazePalm rejects a crop, each of the 21 predicted keypoints is assigned a worst-case error of 256 px before computing PCK, AUC, and MPJPE. This convention is strictly more conservative than published MediaPipe benchmarks that score only successfully detected frames.

B. Experimental Setup and Training Behavior

Table III summarizes the design improvements introduced in V2 to address the overfitting and instability observed in V1. The V2 changes—BlazeHandLandmark initialization, structured augmentation, the Wing+Bone composite loss, cosine LR decay, periodic validation, and gradient clipping during fine-tuning—constitute the configuration evaluated throughout the remainder of this paper. Fig. 3 shows V2 validation metrics over 100 distillation epochs, evaluated against teacher-decoded coordinates on a 200-sample RHD2D subset every 5 epochs. PCK@0.2 peaks at epoch 65 (0.769), but AUC and MPJPE continue to improve through epoch 100 with no regression in any metric. Epoch 100 is therefore selected as the Stage 1 checkpoint: the model continues refining spatial calibration after the coarse-threshold PCK has saturated, validating the cosine LR schedule that keeps the optimizer making fine adjustments in the low-LR regime.

C. Quantitative Results

Results are reported in two tiers reflecting the distinct roles of each model. Table IV reports all models trained or fine-tuned on RHD2D, evaluated on the full 2,727-sample test set against GT annotations. Table V reports the pre-training baseline references that contextualize the student architecture’s initialization state *prior* to distillation alongside the MediaPipe Hands SDK domain-gap reference.

The fine-tuned student (FT V2, ep.150) achieves PCK@0.2=0.748 and AUC=0.723, improving over the

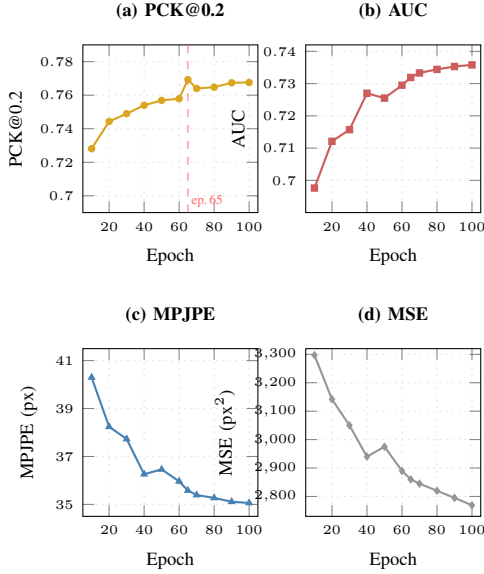


Fig. 3. Distillation V2 validation metrics over 100 epochs. PCK@0.2 peaks at epoch 65 while AUC and MPJPE continue improving through epoch 100.

TABLE IV
TRAINED MODEL COMPARISON — FULL RHD2D TEST SET (2,727 SAMPLES, VS. GT ANNOTATIONS)

Model	Params (M)	PCK@0.2	AUC	MPJPE (px)	Det. Rate	GPU FPS	Δ AUC vs. Direct
Direct Sup. (ep. 25)	2.01	0.745	0.686	40.34	99.96%	139.0	ref.
Stage 1: Dist. V2 (ep. 100)	2.01	0.746	0.721	37.23	100%	139.0	+0.035
Stage 2: FT V2 (ep. 150)	2.01	0.748	0.723	36.87	100%	139.0	+0.037
Teacher (HRNetV2-W18) ^a	9.64	0.992	0.902	2.21	100%	6.9	—

^aUpper-bound reference (official MMPose benchmark); not a deployment candidate.

direct supervision baseline by +3.7pp AUC and −3.47px MPJPE at near-identical PCK@0.2. This confirms the central claim of the pipeline: teacher-decoded sub-pixel targets carry richer spatial information than hard GT labels. Stage 2 fine-tuning contributes consistent but modest gains over Stage 1 (+0.002 AUC, −0.36px MPJPE), indicating that Phase 1 distillation captures the bulk of transferable knowledge while fine-tuning serves as a convergence refinement. The HRNetV2-W18 teacher is reported as an upper-bound reference only; at 6.9FPS it is not a deployment candidate. Notably, direct supervision reaches its best checkpoint at epoch 25 of 100 and then degrades, whereas the distilled models continue improving through their full training horizon without overfitting. Table VI consolidates accuracy, efficiency, and open-source trainability for all models considered in this study.

D. Efficiency and Real-Time Performance

The fine-tuned student delivers 4.8× fewer parameters and 4.8× smaller weights than the HRNetV2-W18 teacher, running at 139.0FPS on GPU—a 20.2× full-pipeline speedup and 4.6× above the 30FPS real-time threshold. CPU throughput (25.6FPS median) falls just below the threshold, representing the primary deployment limitation on CPU-only edge devices.

TABLE V
PRE-TRAINING BASELINE REFERENCE — INITIALIZATION STATES AND DOMAIN-GAP REFERENCE (2,727 SAMPLES[†], VS. GT)

Model	PCK@0.2	AUC	MPJPE (px)	Det. Rate
Untrained BlazeHandLandmark (random init.)	0.005	0.025	186.96	100%
BlazeHandLandmark (zmurec, pre-distillation weights)	0.301	0.395	104.29	80.05%
MediaPipe Hands SDK (domain-gap ref.) [‡]	0.223	0.307	133.27	66.23%
FT V2 ep. 150 (after distillation)	0.748	0.723	36.87	100%

[†]Full 2,727-sample test set; detection failures scored at 256px per joint (not excluded).

[‡]MediaPipe Hands SDK (model_complexity=1, 1.98 M params) run as a full pipeline on pre-cropped inputs; domain-gap artifact only — see Sec. V-A.

TABLE VI
CONSOLIDATED MODEL COMPARISON

Criterion	MP SDK ^c	BlazeHand (zmurec)	Teacher HRNetV2-W18	Direct Sup.	Dist. V2-100	FT V2 -150
Params.	1.98M	2.01M	9.64M	2.01M	2.01M	2.01M
GPU FPS	— ^c	— ^c	6.9	139	139	139
PCK@0.2 ^a	0.223	0.301	0.992 ^b	0.745	0.746	0.748
AUC ^a	0.307	0.395	0.902 ^b	0.686	0.721	0.723
MPJPE ^a (px)	133.27	104.29	2.21 ^b	40.34	37.23	36.87
Det. rate	66.23%	80.05%	100%	99.96%	100%	100%
RT (GPU)	Yes ^c	Yes ^c	No	Yes	Yes	Yes
Open/retrain	No	No	Yes	Yes	Yes	Yes
KD supervision	No	No	—	No	Yes	Yes

^aEvaluated on full 2,727-sample RHD2D test set vs. GT annotations; detection failures scored at 256px per joint.

^bOfficial MMPose benchmark on RHD2D.

^cMP SDK and BlazeHand (zmurec) run as inference-only baselines on pre-cropped inputs; GPU FPS not benchmarked. Mobile GPU: 5.3–16.1ms [8]; not comparable to RTX 4050.

TABLE VII
MULTI-BATCH GPU THROUGHPUT — STUDENT (FT V2 EP. 150)

Batch Size	Throughput (img/s)
1	126.3
4	473.1
8	891.9
16	905.1

Batch throughput scales near-linearly to batch 8 (892img/s) before plateauing at batch 16 (905img/s), confirming efficient GPU utilization for batched video pipelines (Table VII). For mobile context, the official MediaPipe Hands Full model reports 5.3–16.1ms on Pixel 3 / S20 / iPhone 11 [8]; this is not directly comparable due to different hardware and training domain, but indicates that the BlazeHandLandmark architecture itself is well-matched to mobile deployment.

E. Qualitative and Per-Joint Results

Table VIII and Fig. 4 report per-joint PCK@0.2 for the three trained models on the full RHD2D test set. The fine-tuned student achieves the best score on 15 of 21 joints, with the PCK range widening from 0.073 in direct supervision to 0.141 in FT V2. MCP joints (Thumb, Index, Ring, Pinky) consistently register the lowest per-joint scores across distilled models, while distal joints (DIP, Tip) register the highest. The structural interpretation of this pattern is discussed in Sec. VI-C. Fig. 5 shows teacher-vs.-student skeleton overlays on six RHD2D test images; the student reproduces the teacher’s skeleton topology across varied poses and partial

TABLE VIII
PER-JOINT PCK@0.2 — TRAINED MODELS (RHD2D, vs. GT)

Joint	Direct Sup.	Dist. V2-100	FT V2-150
Wrist	0.742	0.801	0.802
Thumb MCP	0.702	0.677	0.678
Thumb PIP	0.742	0.743	0.741
Thumb DIP	0.756	0.755	0.754
Thumb Tip	0.769	0.801	0.805
Index MCP	0.719	0.666	0.665
Index PIP	0.749	0.754	0.750
Index DIP	0.763	0.759	0.759
Index Tip	0.773	0.750	0.755
Middle MCP	0.728	0.720	0.720
Middle PIP	0.763	0.781	0.786
Middle DIP	0.766	0.790	0.797
Middle Tip	0.775	0.779	0.777
Ring MCP	0.713	0.696	0.693
Ring PIP	0.756	0.784	0.785
Ring DIP	0.751	0.784	0.789
Ring Tip	0.759	0.802	0.806
Pinky MCP	0.716	0.667	0.673
Pinky PIP	0.726	0.707	0.712
Pinky DIP	0.733	0.725	0.730
Pinky Tip	0.736	0.726	0.732
Mean	0.745	0.746	0.748
Range (max—min)	0.073	0.137	0.141

Bold = best trained model per joint.

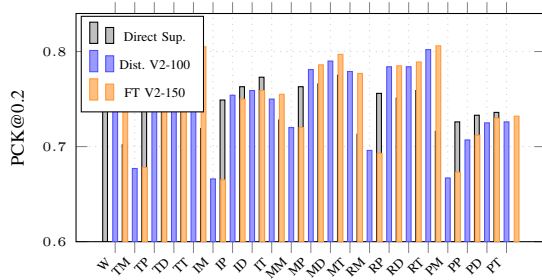


Fig. 4. Per-joint PCK@0.2. X-axis: W = Wrist; T/I/M/R/P = Thumb/Index/Middle/Ring/Pinky; M/P/D/T = MCP/PIP/DIP/Tip. Distilled models (blue/orange) show the proximal-to-distal gradient; direct supervision (gray) is nearly flat.

occlusions, with visible deviations clustering at MCP joints, consistent with the quantitative findings above. Fig. 6 shows live webcam inference on the researcher’s own hand—a real-world capture entirely outside the synthetic RHD2D domain—where the model produces stable predictions without any domain-specific adaptation, indicating that the learned representations generalize beyond the training distribution.

VI. ANALYSIS OF RESULTS

A. Heterogeneous KD Bridges the Representation Gap

The central technical contribution of this work is the use of the MSRAHeatmap codec as a bridge between fundamentally different output modalities: the heatmaps produced by the HRNetV2-W18 teacher and the direct coordinate regression of the BlazeHandLandmark student. Decoding teacher heatmaps into sub-pixel coordinate targets at every training step supervises both models in a common 2D pixel space without architectural modification. The resulting student achieves

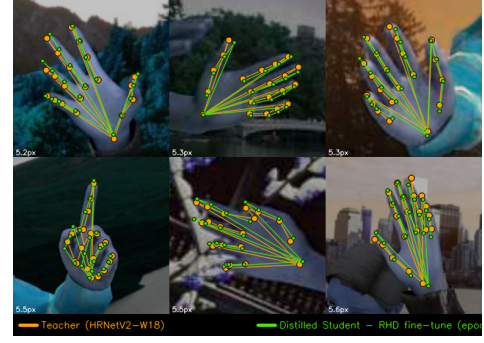


Fig. 5. HRNetV2-W18 teacher (• orange) vs. fine-tuned student (• green) on six RHD2D samples. Per-sample EPE (px) at bottom-left.

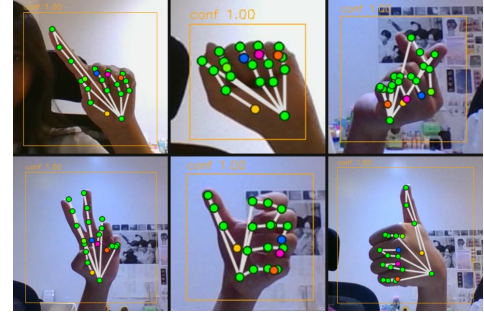


Fig. 6. Live webcam inference. All 21 keypoints predicted in real time from a single RGB frame on out-of-distribution real-hand input.

PCK@0.2=0.748 with 100% detection on RHD2D, confirming that heterogeneous response-based distillation is a viable strategy for cross-architecture pose compression.

B. Distillation Improves Spatial Calibration

At nearly identical PCK@0.2, the distilled student improves AUC by +3.7 pp and reduces MPJPE by 3.47 px relative to direct supervision on ground-truth labels. The AUC gain is the load-bearing result: it measures performance at stricter thresholds (5–20 px) where the lenient 51.2 px PCK@0.2 window is insensitive. The student also exhibits a wider per-joint PCK range (0.141 vs. 0.073 for direct supervision), confirming that it inherits the teacher’s confidence distribution rather than the uniform weighting implicit in GT labels.

C. Per-Joint Structural Patterns

Table VIII reveals two structural patterns. First, distilled models develop a per-joint specialization that direct supervision lacks: the PCK range widens from 0.073 (Direct Sup.) to 0.141 (FT V2), as the distilled student inherits the teacher’s confidence distribution rather than treating all GT labels with equal weight. Second, a clean proximal-to-distal gradient emerges: MCP joints (PCK 0.665–0.720, MPJPE 41–48 px) consistently lag behind PIP (0.707–0.786, 34–41 px), DIP (0.725–0.797, 32–39 px), and Tip joints (0.726–0.806, 31–38 px). This reflects the teacher’s own confidence distribution: MCP joints are frequently occluded by the palm, producing broader heatmap peaks and noisier decoded coordinate targets,

whereas distal joints retain distinctive local appearance that the student’s depthwise-separable feature extractor can reliably exploit.

D. An Open, Retractable Alternative to MediaPipe Hands

Starting from the BlazeHandLandmark (zmurez) pre-distillation weights, the proposed pipeline raises RHD2D PCK@0.2 from 0.301 to 0.748 and raises the detection rate from 80.05% to 100%, eliminating the 19.95% detection-failure mode of the zmurez initialization on synthetic data. For comparison, the MediaPipe Hands SDK, which runs as a full two-stage pipeline, achieves only PCK@0.2=0.223 with a 66.23% detection rate on the same test set, reflecting the additional failure mode introduced by the BlazePalm detector on RHD2D’s flat-shaded synthetic imagery. Neither result is a deficiency of MediaPipe itself: the official SDK was trained on real-world and photorealistic-synthetic imagery [8], which differs substantially from the flat-shaded RHD2D render style. The contribution here is that the same student architecture, when retrained through the proposed pipeline, can be adapted to an arbitrary target domain without proprietary data or closed training infrastructure, addressing the reproducibility and adaptability limitations that motivate this work.

E. Real-Time Efficiency with Significant Compression

The student delivers $4.8\times$ parameter compression (2.01 M vs. 9.64 M) and a $20.2\times$ GPU full-pipeline speedup over the teacher, reaching 139 FPS on an RTX 4050 laptop GPU, which is well above the 30 FPS real-time threshold. CPU throughput (25.6 FPS) remains the principal limitation for edge-only deployments without GPU support and is the natural target for future quantization or mobile-runtime export.

VII. CONCLUSIONS AND RECOMMENDATIONS

This work proposed and validated a heterogeneous knowledge distillation framework for 2D hand pose estimation that bridges a heatmap-based teacher (HRNetV2-W18 with DARK decoding) and a regression-based student (BlazeHandLandmark) through MSRAHeatmap codec decoding. The framework requires no architectural modification to either model and produces a fully open and retrainable pipeline.

The results satisfy the three objectives of the problem statement. First, the heterogeneous KD bridge transfers the spatial reasoning of a teacher nearly $5\times$ larger and $20\times$ slower into a student that remains fully deployable at real-time speeds. Second, retraining through this pipeline yields an open alternative to MediaPipe Hands that achieves 100% detection and PCK@0.2=0.748 on the RHD2D synthetic domain, where the original SDK fails. Third, the framework is superior to naive direct supervision: the teacher’s soft targets produce measurably better spatial precision under stricter evaluation thresholds.

Per-joint analysis further reveals a clear proximal-to-distal performance gradient that reflects the teacher’s own confidence structure: distal joints benefit most from sub-pixel heatmap

supervision, while MCP joints, frequently occluded and producing broader teacher heatmaps, remain the weakest points. This structure is an inherent property of the heterogeneous KD paradigm and motivates the directions outlined below.

Future work should prioritize modulating the per-joint distillation loss by teacher heatmap peak sharpness to reduce noise at occluded MCP joints, and extending evaluation to real-world benchmarks such as OneHand10K (which could not be procured in time due to private gating), FreiHAND, and HO-3D. Integrating a lightweight palm detector such as BlazePalm [8] would complete a fully deployable end-to-end pipeline comparable to MediaPipe Hands, while post-training quantization and mobile-runtime export (TFLite, ONNX Runtime, CoreML) would close the gap between the student’s GPU and CPU throughput. Finally, because the pipeline is fully open source, it can be retargeted to specialized domains—surgical tracking, domain-specific sign language, and industrial safety—where closed commercial solutions cannot currently be retrained.

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APPENDIX A EXPERIMENTAL ENVIRONMENT

Table IX and Table X detail the hardware and software configurations used in this study.

TABLE IX
SYSTEM ENVIRONMENT

Component	Specification
OS / Python	Windows 11 / 3.10
Frameworks	PyTorch 2.1.0+cu121 / MMPose 1.3.2
CUDA	12.1
GPU	NVIDIA GeForce RTX 4050 Laptop GPU

TABLE X
HYPERPARAMETER CONFIGURATION

Parameter	Phase 1	Phase 2	Baseline
Optimizer	Adam	Adam	Adam
Learning Rate	1×10^{-4}	5×10^{-5}	1×10^{-4}
LR Scheduler	Cosine	Cosine	Cosine
Batch Size	32	32	32
Total Epochs	100	50	100
Grad. Clip	—	5.0 (norm)	—
Wing Loss (w, ϵ)	10.0, 2.0	10.0, 2.0	10.0, 2.0
Bone Loss λ	0.002	0.002	—

APPENDIX B ARCHITECTURE DETAILS

Teacher (HRNetV2-W18 + DARK): Implemented via MMPose as a TopdownPoseEstimator with the HRNetV2-W18 backbone (channel widths 18, 36, 72, 144 across four stages), a FeatureMapProcessor neck that concatenates all four branch feature maps into 270 channels, and a HeatmapHead producing $21 \times 64 \times 64$ heatmaps. Heatmaps are decoded via the MSRAHeatmap codec using DARK sub-pixel Taylor expansion ($\sigma = 2$). The model contains 9.6417M parameters (checkpoint size 37.15 MB).

Student (BlazeHandLandmark): A lightweight BlazeBlock-based regression network initialized from MediaPipe pretrained weights. The backbone progresses through four stages (24, 48, 96, 96 channels) with FPN-style feature fusion, followed by a 29-block deep refinement module yielding a $(B, 288, 1, 1)$ bottleneck. Three output heads produce `hand_flag` (detection confidence $\in [0, 1]$), `handed` (handedness, unused in distillation), and `landmarks` (shape $B \times 21 \times 3$, normalized to $[0, 1]$ by dividing by 256). For distillation and evaluation, `landmarks[:, :, : 2] \times 256` yields pixel-space (x, y) coordinates of shape $(B, 21, 2)$. The model contains 2.0139M parameters (checkpoint size 7.75 MB on disk). Thesis checkpoint: `distilled_v2_ft_epoch_150.pth` (100 epochs distillation + 50 epochs fine-tuning).

APPENDIX C PERFORMANCE METRICS

This appendix defines all quantitative metrics used to evaluate the teacher, student, and baseline models. Unless otherwise stated, metrics are computed over the full RHD2D test set (2,727 images) with 21 keypoints per sample.

A. Percentage of Correct Keypoints (PCK@0.2)

PCK@0.2 is the primary accuracy metric for 2D hand pose estimation. A predicted keypoint $\hat{p}_k$ is *correct* if its Euclidean distance to the ground-truth location p_k^* falls within a threshold τ :

$$\text{PCK}@ \alpha = \frac{1}{KN} \sum_{n=1}^N \sum_{k=1}^K \mathbf{1} \left[\|\hat{p}_k^{(n)} - p_k^{*(n)}\|_2 \leq \tau \right], \quad (5)$$

where $\tau = \alpha \times \text{reference_length}$. In this study, the reference length is the input image size (256 px), $\alpha = 0.2$, giving a fixed pixel threshold of $\tau = 51.2$ px. PCK@0.2 is reported as a proportion in $[0, 1]$.

Interpretation: The generous threshold makes PCK@0.2 most sensitive to catastrophic failures (missed joints, large structural errors). Two models with similar PCK@0.2 may differ substantially in fine-grained precision, which is why AUC and MPJPE are reported alongside it.

B. Area Under the PCK Curve (AUC)

AUC integrates the PCK score over a range of distance thresholds:

$$\text{AUC} = \frac{1}{M} \sum_{m=1}^M \text{PCK}(\tau_m), \quad \tau_m \in [0, 128] \text{ px}, \quad M = 20, \quad (6)$$

with the 20 thresholds uniformly spaced from 0 to 128 px. AUC is reported in $[0, 1]$.

Interpretation: AUC is the most discriminative aggregate metric in this study. Because it includes strict thresholds (e.g., 5–20 px), it rewards models that place keypoints *close* to ground truth, not merely within the generous 51.2 px PCK window. An AUC improvement without a corresponding PCK@0.2 improvement indicates better spatial precision at tight tolerances—the primary signal used to demonstrate the benefit of KD over direct supervision.

C. End-Point Error / Mean Per-Joint Position Error (EPE / MPJPE)

EPE and MPJPE denote the same quantity here: the mean Euclidean distance (px) between predicted and ground-truth keypoints, averaged over all joints and samples:

$$\text{EPE} = \text{MPJPE} = \frac{1}{KN} \sum_{n=1}^N \sum_{k=1}^K \|\hat{p}_k^{(n)} - p_k^{*(n)}\|_2. \quad (7)$$

EPE is expressed in pixels on the 256×256 input space; lower is better. EPE is the term used in model-zoo benchmarks (MMPose, MediaPipe paper), while MPJPE is used in the training validation loop and per-joint tables.

D. Mean Squared Error (MSE)

MSE is the mean squared Euclidean distance between predicted and ground-truth coordinates:

$$\text{MSE} = \frac{1}{KN} \sum_{n=1}^N \sum_{k=1}^K \|\hat{p}_k^{(n)} - p_k^{*(n)}\|_2^2. \quad (8)$$

MSE is reported in px^2 . Because squaring amplifies large errors, MSE is more sensitive to outliers than MPJPE.

Interpretation: A model may have lower MSE than a competitor while having higher MPJPE if it avoids rare catastrophic misplacements but is otherwise less precise. In this study, the direct supervision baseline achieves lower MSE ($\approx 2,061 \text{ px}^2$) than the distilled models ($\approx 3,059 \text{ px}^2$) because its training objective directly minimizes MSE against ground-truth coordinates. However, its lower AUC (0.686 vs. 0.723) reveals that this MSE advantage is an artifact of objective alignment, not genuine spatial improvement.

E. Detection Rate

The detection rate is the fraction of test samples for which the student outputs `hand_flag` ≥ 0.5 :

$$\text{Detection Rate} = \frac{|\{n : \text{hand_flag}^{(n)} \geq 0.5\}|}{N}. \quad (9)$$

Interpretation: A rate below 100% means the model is uncertain about hand presence in some crops, even though each RHD2D test image contains exactly one hand. The distilled thesis model achieves a detection rate of 100% (2,727/2,727); the direct supervision baseline achieves 99.96% (2,726/2,727).

F. Summary of Metrics

Table XI provides a concise reference for all metrics used in this study.

TABLE XI
SUMMARY OF EVALUATION METRICS

Metric	Full Name	Units	Better	Threshold / Range
PCK@0.2	Percentage of Correct Keypoints	proportion	↑	$\tau = 51.2 \text{ px}$
AUC	Area Under PCK Curve	proportion	↑	$\tau \in [0, 128] \text{ px}$
EPE / MPJPE	End-Point / Mean Per-Joint Error	px	↓	—
MSE	Mean Squared Error	px^2	↓	—
Det. Rate	Detection Rate (<code>hand_flag</code> ≥ 0.5)	%	↑	—

APPENDIX D DETAILED NUMERICAL RESULTS

Table XII provides a focused per-joint comparison between the distilled thesis model (`distilled_v2_ft_epoch_150.pth`) and the direct supervision baseline on key joints. The full 21-joint comparison appears in Table VIII.

TABLE XII
PER-JOINT RESULTS: DISTILLED STUDENT VS. DIRECT SUPERVISION
BASELINE (RHD2D TEST)

Joint	Direct Baseline		Distilled Student	
	PCK@0.2	MPJPE (px)	PCK@0.2	MPJPE (px)
Wrist	0.742	40.12	0.802	33.35
Thumb Tip	0.769	39.35	0.805	31.67
Index Tip	0.773	39.20	0.755	35.21
Middle Tip	0.775	39.34	0.777	32.32
Ring Tip	0.759	39.79	0.806	31.03
Pinky Tip	0.736	40.81	0.732	37.65
Mean	0.745	40.34	0.748	36.87

APPENDIX E PHASE 1 VALIDATION PROGRESS

Table XIII logs validation metrics recorded on a 200-sample held-out subset every 20 epochs during Phase 1 distillation, demonstrating convergence characteristics. Metrics here measure student–teacher agreement (student predictions vs. teacher-decoded coordinates), not ground-truth accuracy.

TABLE XIII
PHASE 1 DISTILLATION VALIDATION METRICS (200-SAMPLE SUBSET, VS. TEACHER-DECODED COORDINATES)

Epoch	PCK@0.2	AUC	MPJPE (px)	MSE (px^2)
20	0.745	0.712	38.30	3,075
40	0.754	0.728	36.30	2,865
60	0.758	0.731	36.00	2,905
80	0.765	0.734	35.30	2,805
100	0.767	0.736	35.06	2,769

APPENDIX F JOINT AND BONE TOPOLOGY

The student predicts 21 keypoints ($k \in [0, 20]$), following the standard MediaPipe hand topology: joint 0 is the wrist; joints 1–4 are the Thumb MCP, PIP, DIP, and Tip; joints 5–8 the Index finger; 9–12 Middle; 13–16 Ring; and 17–20 Pinky.

The Bone Loss $\mathcal{L}_{\text{bone}}$ is computed over the set of 20 skeletal connections $\mathcal{B}$ as the mean squared difference between predicted and teacher-decoded bone lengths:

$$\mathcal{L}_{\text{bone}} = \frac{1}{|\mathcal{B}|} \sum_{(i,j) \in \mathcal{B}} (\|\hat{p}_i - \hat{p}_j\|_2 - \|\tilde{p}_i - \tilde{p}_j\|_2)^2, \quad (10)$$

where $\hat{p}_k$ denotes the student’s predicted pixel-space coordinates and $\tilde{p}_k$ denotes the teacher-decoded coordinates produced by the MSRAHeatmap codec at each training step. Ground-truth annotations are *not* used in this loss; the teacher is the sole reference. Connections follow the four-joint chain of each finger (MCP→PIP→DIP→Tip) plus the wrist-to-MCP link for each finger, totaling 20 bone segments: (0, 1), (1, 2), (2, 3), (3, 4) for the thumb, and the analogous four-link chains for Index, Middle, Ring, and Pinky.